

Part XII. Compare the Documented and Artificial Mutations

Item	Wildtype	Documented Mutation
CDS length	5,934 bp	5,934 bp
Protein length	1,977 aa	1,977 aa
Mutation type	None	Single-nucleotide substitution
Reading frame changed	No	No
Premature stop codon appeared	No	No
Amino acids affected	None	1 amino acid (p.I892M)
Expected functional	Normal channel gating and function	Gain of function causing neuronal hyper excitability and Inherited Erythromelalgia (IEM)

This shows a normal, unaltered gene (Wildtype) to a single-nucleotide substitution mutation. The mutation swaps a single DNA letter, which alters just one amino acid in the resulting protein without changing the gene's overall length or reading frame. This small change creates a gain-of-function effect that makes neurons hyper-excitabile, leading to a condition called Inherited Erythromelalgia (IEM).